

$ident, x$		
exp, e	$::=$	
		x
		$()$
		(exp, exp')
		function $x : type \rightarrow exp$ bind x in exp
		$exp\ exp'$
		$\{exp/x\}exp'$ M
		(exp) M
$type, t$	$::=$	
		unit
		$type * type'$
		$type \rightarrow type'$
E	$::=$	
		empty
		$E, x : t$
		$E, x : t, E'$
val, v	$::=$	
		x
		$()$
		(val, val')
		function $x : t \rightarrow exp$
$formula$	$::=$	
		$judgement$
		not ($formula$)
		x in dom (E) M
$terminals$	$::=$	
		$\rightarrow$
		$\vdash$
		$\longrightarrow$
$Jtype$	$::=$	
		$E \vdash \mathbf{ok}$ M
		$E \vdash e : t$
Jop	$::=$	
		$e \longrightarrow e'$
$judgement$	$::=$	
		$Jtype$
		Jop
$user_syntax$	$::=$	
		$ident$
		exp

$|$ *type*
 $|$ *E*
 $|$ *val*
 $|$ *formula*
 $|$ *terminals*

$E \vdash \mathbf{ok}$

$$\begin{array}{c}
\frac{}{\mathbf{empty} \vdash \mathbf{ok}} \quad \text{EOK_EMPTY} \\
\frac{E \vdash \mathbf{ok} \quad \mathbf{not}(x \text{ in dom}(E))}{E, x : t \vdash \mathbf{ok}} \quad \text{EOK_IDENT}
\end{array}$$

$E \vdash e : t$

$$\begin{array}{c}
\frac{E, x : t, E' \vdash \mathbf{ok}}{E, x : t, E' \vdash x : t} \quad \text{EET_IDENT} \\
\frac{E \vdash \mathbf{ok}}{E \vdash () : \mathbf{unit}} \quad \text{EET_UNIT} \\
\frac{E \vdash e_1 : t_1 \quad E \vdash e_2 : t_2}{E \vdash (e_1, e_2) : t_1 * t_2} \quad \text{EET_PAIR} \\
\frac{E, x : t_1 \vdash e : t_2}{E \vdash \mathbf{function } x : t_1 \rightarrow e : t_1 \rightarrow t_2} \quad \text{EET_FUN} \\
\frac{E \vdash e : t_1 \rightarrow t_2 \quad E \vdash e' : t_1}{E \vdash e e' : t_2} \quad \text{EET_APP}
\end{array}$$

$e \longrightarrow e'$

$$\begin{array}{c}
\frac{}{(\mathbf{function } x : t \rightarrow e) v \longrightarrow \{v/x\}e} \quad \text{JO_RED_FUN} \\
\frac{e_1 \longrightarrow e'_1}{e_1 e_2 \longrightarrow e'_1 e_2} \quad \text{JO_RED_CONTEXT1} \\
\frac{e_2 \longrightarrow e'_2}{v_1 e_2 \longrightarrow v_1 e'_2} \quad \text{JO_RED_CONTEXT2} \\
\frac{e_1 \longrightarrow e'_1}{(e_1, e_2) \longrightarrow (e'_1, e_2)} \quad \text{JO_RED_CONTEXT3} \\
\frac{e_2 \longrightarrow e'_2}{(v_1, e_2) \longrightarrow (v_1, e'_2)} \quad \text{JO_RED_CONTEXT4}
\end{array}$$

Definition rules: 12 good 0 bad
Definition rule clauses: 25 good 0 bad